The Impact of Air Pollution, Vegetation Density, and Climate Conditions on Respiratory-Related Mortality in the United States

Jaidan Bhakta, Shubhangi Kumar, Oz Bennett

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1 Abstract

Respiratory-related health issues are increasingly becoming widespread throughout the United States. Chronic Obstructive Pulmonary Disease (COPD) is the fourth-leading cause of death in the U.S. and is projected to rise in the ranks in the near future. There are a plethora of important factors that, taken together, contribute to an explanation of that startling fact. A full explanation, however, is incomplete without an analysis concerning the extent to which different regions across the United States are impacted. Therefore, in this paper, we seek to examine specifically how environmental factors may help explain regional disparities in respiratory-related mortality rates. Particularly, we look to environmental variables like air pollution, loss of vegetation density, and climate conditions, all of which are region-specific and can help explain regional differences. Through an exploratory data analysis, we examine geographic heat maps of the United States to contextualize the correlative relationships between the aforementioned environmental factors and respiratory-related crude mortality rates. Further, we apply machine learning models, specifically Random Forest and Gradient Boosting algorithms which capture non-linear relationships, in order to determine which model best predicts our target variable: crude mortality rate. We examine the performance metrics of each model within each of the four Census regions: Northeast, Midwest, South, and West. Ultimately, we learned that the Random Forest model outperforms the Gradient Boosting model in each region, but the former is not without its flaws. We also applied a risk classification model through a Random Forest model, categorizing regions into high-risk and low-risk. The model classified the Midwest and South as high-risk and the Northeast and West as low-risk, which is entirely consistent with prior research and literature as well as our own exploratory data analysis. Ultimately, we acknowledge that our analysis in this report is not at all determinative of any finding of a causal connection between environmental factors and respiratory-related mortality. After all, there are thousands of confounding variables that impact respiratory health and mortality. Nevertheless, our analysis provides meaningful insight into the predictive analysis of respiratory-related mortality as well as the non-trivial relationships between regional crude mortality rates and air pollution, vegetation density, and climate conditions.

2 Introduction

Today, it is crucial to view our atmosphere as a dynamic system, where hundreds of distinct variables converge and interact. Operating within this dynamic system will allow us to optimally analyze important environmental factors that contribute to respiratory illness in the United States. Chronic Obstructive Pulmonary Disease (COPD), rhinitis, and respiratory infections are particularly pertinent respiratory conditions throughout the US, with COPD being the fourth-leading cause of death in the US and projected to be the third-leading cause in the future [4]. However, not all regions in the US are treated equally. Different regions in the US may experience varying health outcomes based on climatic conditions, vegetation coverage, and air pollutant concentrations.

Air pollution remains a significant environmental and public health concern in the United States, contributing to a wide range of adverse health outcomes across different regions. Emissions from transportation, industrial activity, power generation, and other sources release pollutants such as fine particulate matter (PM2.5), nitrogen dioxide (NO2), and ozone (O3) into the atmosphere [14]. While CO2 is primarily associated with climate change, it also serves as an indicator of combustion-related activities that generate co-pollutants known to affect respiratory health [1]. Exposure to these pollutants has been linked to inflammation of the respiratory system, reduced lung function, and increased susceptibility to respiratory diseases like COPD [10, 14].

PM2.5, NO2, and O3 are associated with both chronic and acute respiratory conditions, with their health impacts shaped by environmental context. Vegetation coverage can help mitigate exposure to pollution through pollutant removal and temperature regulation [9, 12, 11], while climatic factors such as temperature and seasonality influence pollutant formation and respiratory risk. These factors vary widely across U.S regions, contributing to geographic differences in pollution exposure and respiratory health outcomes. The goal of this project is to examine how air pollutants, vegetation coverage, and climatic conditions jointly influence respiratory outcomes, including COPD, rhinitis, and respiratory infections, across the United States.

Respiratory illness remains a major public health concern in the United States, with conditions as COPD, rhinitis, and respiratory infections contributing substantially to morbidity and mortality. Existing research has demonstrated that exposure to air pollutants, including, PM2.5, NO2, and O3, plays a significant role in worsening respiratory outcomes, while climatic factors such as temperature extremes, wildfires, and seasonal variability further amplify these risks[7, 6, 11, 13, 14].A growing body of literature also suggests that environmental context, particularly vegetation coverage, influences air quality, thermal conditions, and population-level exposure to respiratory stressors. However, much of the existing research has examined these factors in isolation, despite evidence that air pollution, vegetation, and climate interact within a dynamic atmospheric system. Large-scale, regionally resolved analyses that jointly consider these environmental variables across the United States remain limited[15, 2, 7]. This study presents an opportunity to address this gap by examining how air pollutants, vegetation density, and climatic conditions together shape respiratory health outcomes across regions, providing insight that may inform future environmental health research and public health policy in a changing climate.

2.1 Central Research Question

How do air pollutants (PM2.5, NO2, and O3), vegetation density, and climatic conditions interact across the United States to shape respiratory health outcomes, specifically respiratory-related mortality?

2.1.1 Sub-Question 1

Can machine learning models accurately predict regional respiratory health outcomes using air pollutant levels, vegetation coverage, and climatic variables?

2.1.2 Sub-Question 2

Which environmental factors (PM2.5, NO2, O3, CO2 emissions, vegetation, climate variables) are the most important predictors of respiratory health outcomes in machine learning models?

2.1.3 Sub-Question 3

Do machine learning models identify nonlinear relationships or exposure thresholds between air pollutants and respiratory health outcomes that linear models do not capture?

2.1.4 Sub-Question 4

Can machine learning models classify U.S. regions into high-risk vs low-risk respiratory health categories based on combined environmental exposures?

3 Methodology

3.1 Data Processing and Integration

Our primary data U.S. Environmental Protection Agency's Air Quality System (AQS) monitoring database which provides detailed annual air quality and meteorological observations at the county and state levels. We restricted our analysis to the years 2000 through 2016 in order to examine long-term environmental trends while avoiding distortions associated with the COVID-19 period. The raw AQS data were provided as separate annual files. These 17 yearly datasets were first programmatically concatenated into a single unified dataset to allow for consistent longitudinal analysis across time.

To achieve greater consistency across the years, we standardized the column names and ensured that the data types were uniform. The measurements of pollutants were kept in their reported units (such as NO2 in parts per billion, O3 in parts per million, and PM2.5 in micrograms per cubic meter). Where applicable, non-numeric entries and placeholder symbols were converted into missing values and treated appropriately. Given that air quality monitoring coverage is inconsistent across all U.S. counties, we limited our state- and county-level summaries to locations that had a sufficient number of recorded observations during the study period. Missing values were not imputed at this point; instead, we conducted aggregation procedures using the available observations to avoid introducing any artificial bias.

Following the data cleaning, we aggregated the information by state and county to determine average pollutant concentrations and climate metrics for the years 2000 to 2016. This aggregation allowed us to produce comparative heat maps and regional summaries while retaining the ability to analyze annual trends. Geographic identifiers were carefully preserved to facilitate future merging with datasets concerning vegetation and respiratory mortality.

Vegetation dataset was obtained from the World Resources Institute and provide county-level measures of annual tree cover loss. Because our air quality analysis spans 2000–2016, the vegetation dataset was truncated to the same time period to ensure temporal consistency. County and state identifiers were standardized to match those in the AQS dataset, allowing for accurate cross-dataset integration. It is important to note that this dataset measures vegetation loss rather than total vegetation density, which may influence interpretation of environmental buffering effects.

Finally, the last dataset involves the respiratory-related crude mortality rates by State and County from CDC Wonder. The crude mortality rate measures the number of respiratory-related deaths per 1,000 people in a given year. This is a crucial dataset for our purposes because we want to analyze the crude mortality rate by region, side-by-side with the regional concentrations of air pollutants. The CDC has provided a comprehensive dataset of respiratory-related mortality rates by region from 2000 to 2016, which is our target year range because we wanted to be as detached as possible from the COVID-19 period.

This project combines environmental and health datasets at the county level throughout the United States. The dataset includes metrics of air pollution (NO2, PM2.5, and Ozone), climate factors (ambient and outdoor temperature, wind speed and direction), vegetation indicators (tree coverage loss), and respiratory health outcomes measured using crude mortality rates.

Data processing involved cleaning column names, addressing missing values, and merging datasets

using county-level identifiers. Rows with missing values in key variables were removed to ensure consistency across analyses. The final dataset for modeling consisted of 1,122 counties.

3.2 Introduction to Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to examine distributions, identify outliers, and uncover regional trends. Visualizations such as histograms and geographic maps revealed substantial variation in environmental exposures and respiratory outcomes across the United States.

In particular, Southern and Midwestern regions exhibited higher average mortality rates compared to the Northeast and West, suggesting geographic disparities in respiratory health risk. Additionally, the distribution of mortality rates was right-skewed, indicating the presence of extremely high-risk counties that may be more difficult to predict.

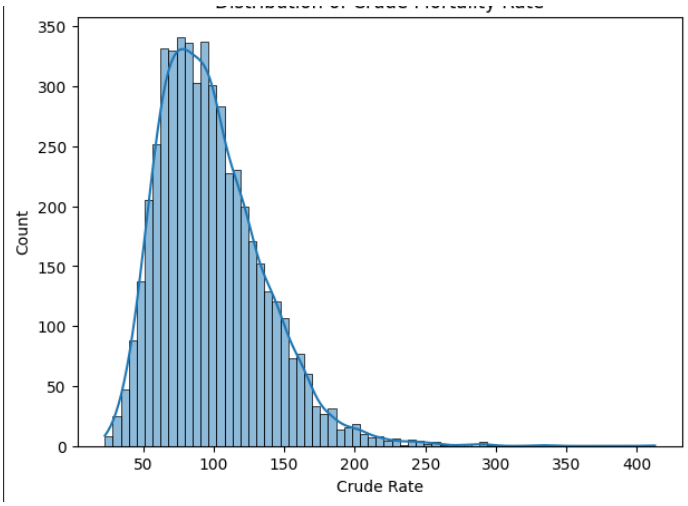


Figure 1: Distribution of Crude Mortality Rates

To better understand the behavior of the response variable, a distribution plot of crude mortality rates was generated. The distribution provides insight into the variability and spread of mortality across counties, highlighting whether the data are skewed or concentrated within a narrow range. Most of the county crude rates seem to be concentrated between 80 and 120, while there are some notable outlier counties near the 250 to 300 range. Understanding the shape of the target variable is important for selecting appropriate modeling techniques and interpreting prediction performance.

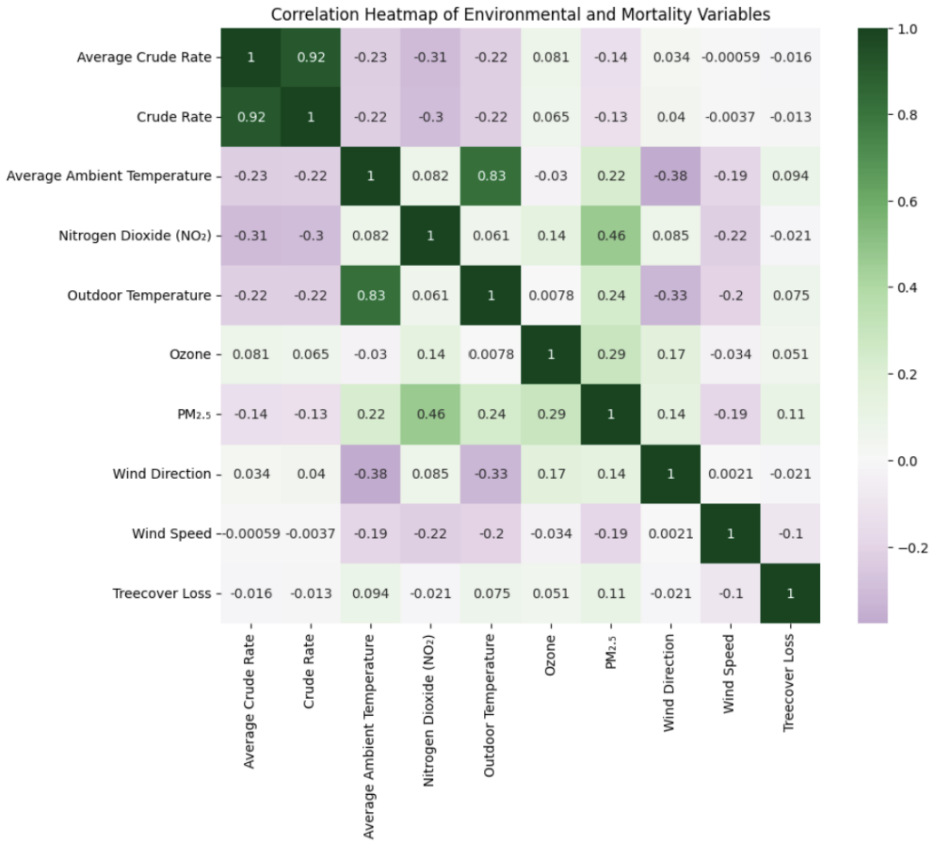


Figure 2: Correlation Heatmap

A correlation heatmap was used to examine relationships between environmental variables and mortality. Interestingly, nitrogen dioxide(NO2) shows a moderate negative correlation with crude mortality rates (-0.39), suggesting that counties with higher NO2 levels tend to have lower mortality. While this is counterintuitive given the known health impacts of air pollution, it likely reflects underlying confounding factors such as urbanization, healthcare access, and socioeconomic differences.

PM2.5 shows little to no correlation with mortality, indicating that its effects may not be captured through simple linear relationships or may be influenced by other interacting variables. Climate variables, including temperature, display weak negative relationships with mortality, while vegetation loss shows a slight positive association. Overall, these findings highlight the complexity of environmental health relationships and reinforce the need for multivariate modeling approaches.

3.3 Introduction to Correlative Analysis and Machine Learning Models

To examine the relationships between environmental variables and respiratory health outcomes, a correlation analysis was carried out. The results revealed that pollution variables, including NO2 and PM2.5, are associated with mortality, although these associations are not strongly linear. Climate and vegetation variables exhibited more complex interactions, emphasizing the importance of multivariate modeling.

To capture nonlinear relationships, machine learning models were implemented. Random Forest and Gradient Boosting were selected due to their ability to model complex interactions between variables. The dataset was split into training (70%), validation (15%), and testing (15%) sets. Model performance was evaluated using metrics including accuracy, balanced accuracy, ROC-AUC,

and F-1 score. Random Forest was expected to outperform other models due to its robustness and ability to handle nonlinear relationships and feature interactions.

3.4 Introduction to Risk Classification

To identify high-risk regions, counties were classified into binary categories based on the 75th percentile of crude mortality rates. Counties at or above this threshold were labeled as high risk, while those below were categorized as low risk. This resulted in an imbalanced dataset, with a larger proportion of counties classified as high risk. A Random Forest classifier with SMOTE was implemented to address class imbalance, and the optimal classification threshold was selected based on maximizing the F1-score for the high-risk class. The final model achieved an AUC of 0.908, indicating strong overall discriminatory performance between high and low-risk counties.

4 EDA Results

4.1 Air Pollution

4.1.1 NO2 Concentrations by State and County



Figure 3: Average NO2 Concentrations by State (parts per billion) from 2000-2016



Figure 4: Average NO2 Concentrations by County (parts per billion) from 2000-2016

It is immediately apparent that there is a stark contrast in NO2 concentrations between several states. For example, in Montana, Wyoming, North Dakota, and South Dakota, we see the lowest levels of NO2 in the nation. On the other hand, in Arizona, Illinois, West Virginia, New Jersey, and

New York, we see the highest concentrations of NO2 in the nation. While further research would be helpful in determining exactly why these drastic differences exist, the preliminary research and general knowledge at our disposal tells us that these observations are not at all unusual. Montana, Wyoming, North Dakota, and South Dakota indeed have the lowest population densities of any other state among the contiguous 48 states (Alaska is the absolute lowest) [8]. However, it is worth noting that Alaska has relatively high levels of NO2. This likely because NO2 concentrations are usually the highest in cooler climates because we typically see increased energy usage for heating during the cooler seasons. Cooler climates also create temperature inversions, trapping NO2 near the ground, preventing them from dispersing [5]. Furthermore, the aforementioned high-NO2 states are also very densely populated, but that fact alone would not be fully sufficient to explain the high concentrations of NO2 in those regions because Florida and Massachusetts are also very densely populated. There are likely other factors like traffic congestion and climate that might contribute to a fuller explanation of this phenomenon.

The county map doesn't seem nearly as helpful as the state map, but it nevertheless provides important insight into how smaller regions differ in NO2 concentrations. It is immediately obvious that large counties like Maricopa County in Arizona and San Bernardino and Los Angeles in California have particularly high levels of NO2 concentration. This is certainly to be expected since those counties are indeed full of traffic congestion and dense populations. For the most part, an analysis of the available counties is rather consistent with what we would expect. While each individual county may not have an unusually high level of NO2, in the aggregate groups of counties can collectively have high concentrations that contribute to their respective state's overall air quality.

4.1.2 PM2.5 Concentrations by State and County



Figure 5: Average PM2.5 Concentrations by State (micrograms/cubic meter) from 2000-2016

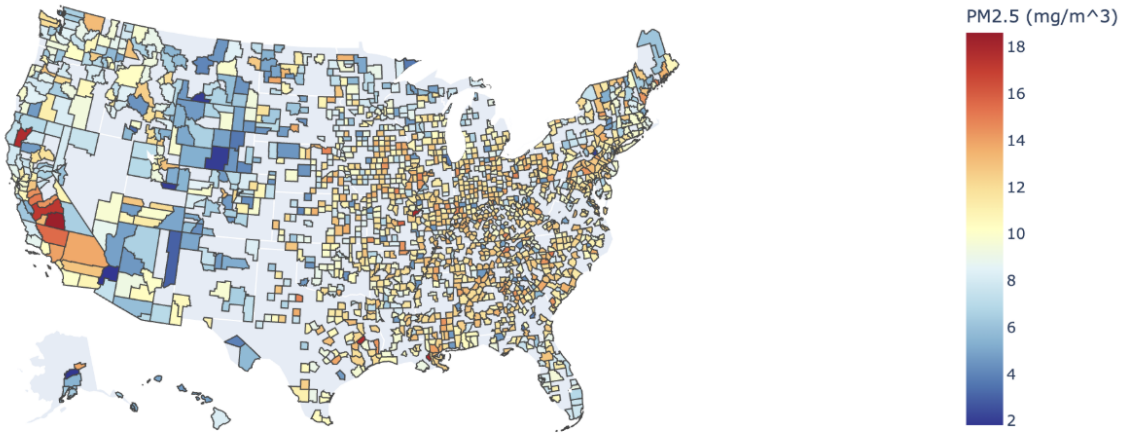


Figure 6: Average PM2.5 Concentrations by County (micrograms/cubic meter) from 2000-2016

Looking at the State map, we already see a vastly different heat map than the NO2 map. The most highly PM2.5-concentrated states include West Virginia, Indiana, Kentucky, Tennessee, Alabama, and Georgia, while the least concentrated states include Wyoming and North Dakota. In the former region, those states are marked by a combination of heavy industrial activity, high traffic density, power plants, and other PM2.5-related emissions. California is an interesting case because as a whole it does not seem to be particularly high in PM2.5 concentration, but when looking at its counties, it is easily apparent that only a handful of densely populated counties account for the highest levels of PM2.5. In Northern California, we see much more mild levels of PM2.5 which balance out the much higher levels we see in the dense south. Comparatively, now looking at the Midwest, South, and Mid-Atlantic regions, we see a much more substantial number of counties that are each relatively high in PM2.5 concentration. Whereas NO2 levels in each county in this region are relatively low, the PM2.5 levels in each county in the same region are relatively high, contributing to the overall PM2.5 concentrations of the states in which they reside.

4.1.3 O3 Concentrations by State and County



Figure 7: Average Ozone Concentrations by State (parts per million) from 2000-2016

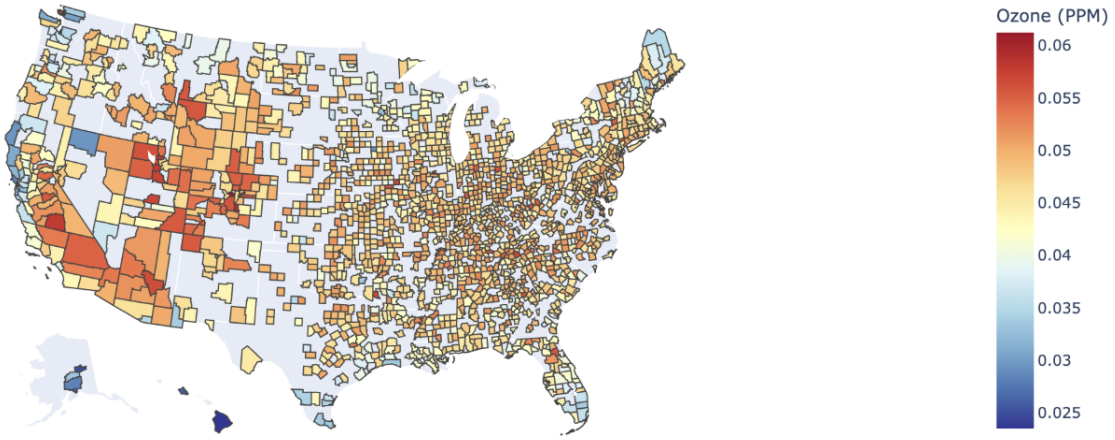


Figure 8: Average Ozone Concentrations by County (parts per million) from 2000-2016

Looking at the State map, it is clear that the states with the highest ozone concentrations include Arizona, Utah, Colorado, Idaho, and Wyoming. We see a similar trend when looking at the county map, where a lot of the counties in the west seem to have relatively high ozone concentrations. These state and county trends are largely consistent with our prior research. In our literature review, we learned that exposure to O3 is usually highest on hot summer days [3]. Certainly, the aforementioned states are drier and have hotter average temperatures than most other states. Notably, Alaska, which seems to have the lowest level of ozone concentration in the United States, has a particularly cool climate, which accounts for it also having a particularly low ozone concentration. It is also worth noting that ozone levels are relatively high in the South and Mid-Atlantic regions, where PM2.5 levels are also incredibly high. Given these facts, we can safely conclude that those regions seems to most prone to poorer air quality than most other states, given that ozone and PM2.5 are both potent and prevalent air pollutants not only in the U.S. but around the world.

4.2 Vegetation Density

4.2.1 Vegetation Density Loss in 2001 by State and County

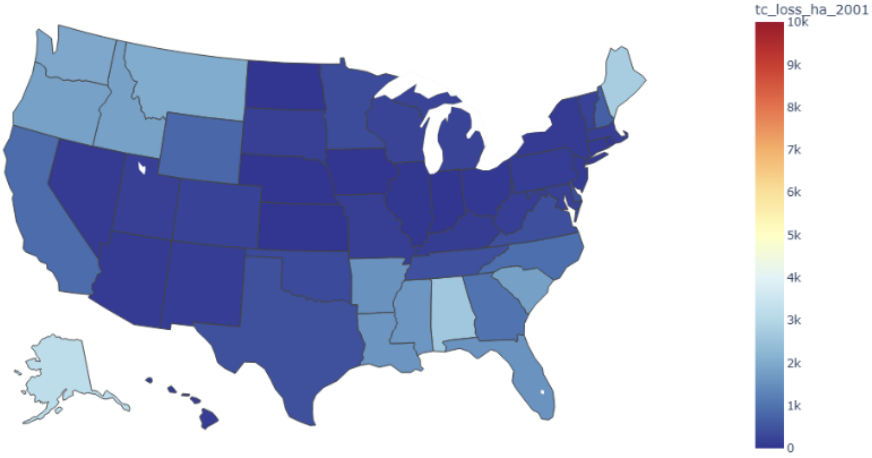


Figure 9: The Loss of Tree Coverage (in hectares) by State in 2001

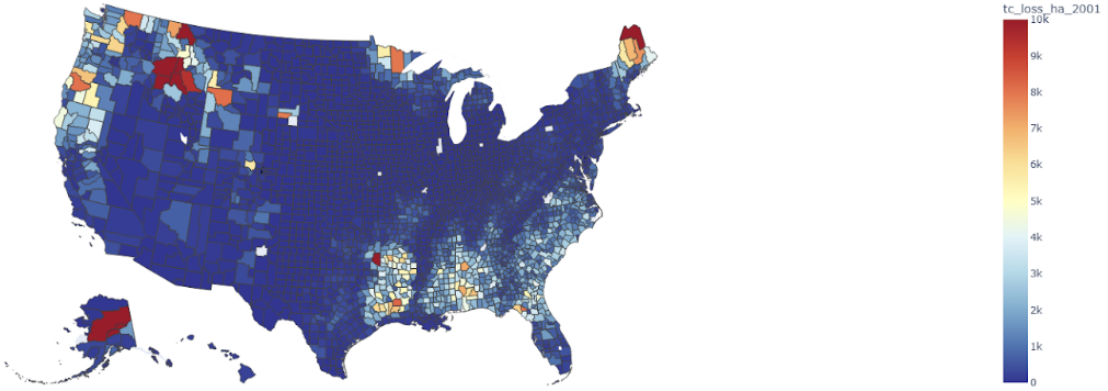


Figure 10: The Loss of Tree Coverage (in hectares) by County in 2001

The 2001 maps illustrates that the distribution of tree cover loss was not uniform across the country. Areas with higher levels of vegetation loss are notably found in the Pacific Northwest, Northern California, certain regions of the Southeast, and Alaska. In contrast, many counties in the Midwest and Northeast display relatively lower levels of tree cover loss during that year.

The pattern of vegetation loss in heavily forested areas is consistent with the dynamics of logging activities, wildfire occurrences, and shifts in land use. In the Southeast, where forestry is economically vital, the heightened vegetation loss likely mirrors the cycles of timber harvesting. In the western states, the decrease in vegetation may be attributed to wildfire activity and environmental stress linked to drought. These regional distinctions are crucial for interpreting pollutant concentrations, as vegetation functions as a natural buffer that can diminish particulate matter buildup and enhance local air quality.

4.2.2 Vegetation Density Loss in 2016 by State and County

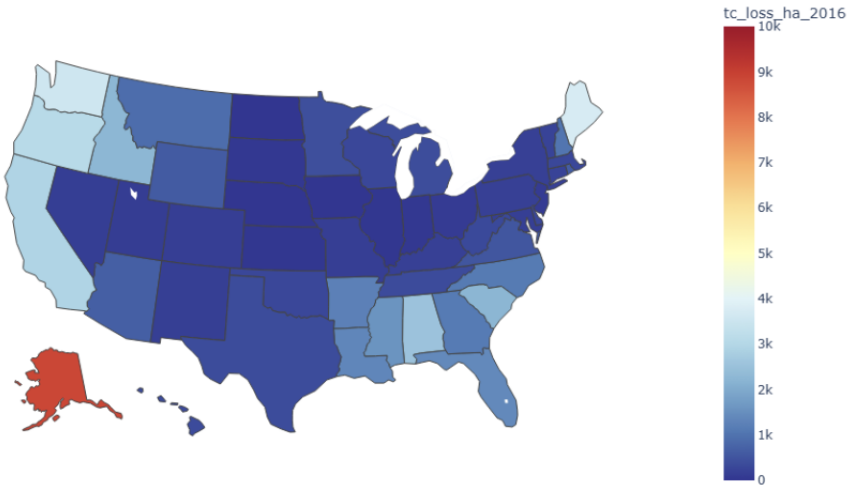


Figure 11: The Loss of Tree Coverage (in hectares) by State in 2016

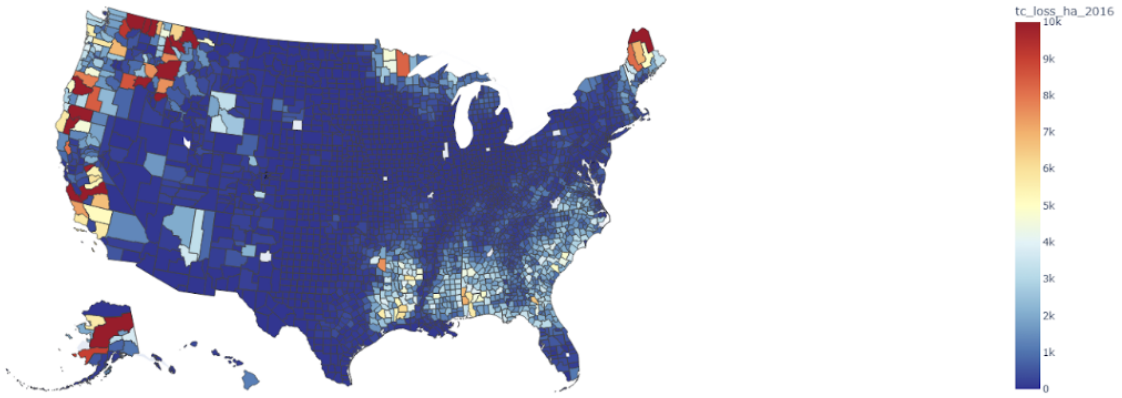


Figure 12: The Loss of Tree Coverage (in hectares) by County in 2016

By 2016, vegetation loss appears more pronounced and geographically widespread in certain western and southeastern regions compared to 2001. Parts of California, the Pacific Northwest, and Alaska show intensified levels of tree cover reduction. Several southeastern counties also continue to demonstrate persistent vegetation loss.

The increased vegetation loss observed in western states may be partially attributable to heightened wildfire activity and prolonged drought conditions during the mid-2010s. Meanwhile, sustained vegetation reduction in the Southeast may reflect continued forestry and land development practices. While vegetation loss does not directly measure air pollution levels, reductions in tree coverage may diminish the environment's capacity to absorb pollutants such as PM2.5 and ozone precursors. This reduction in natural filtration could contribute to greater pollutant persistence in affected regions.

4.3 Climate Conditions

4.3.1 Temperature Trends by State and County

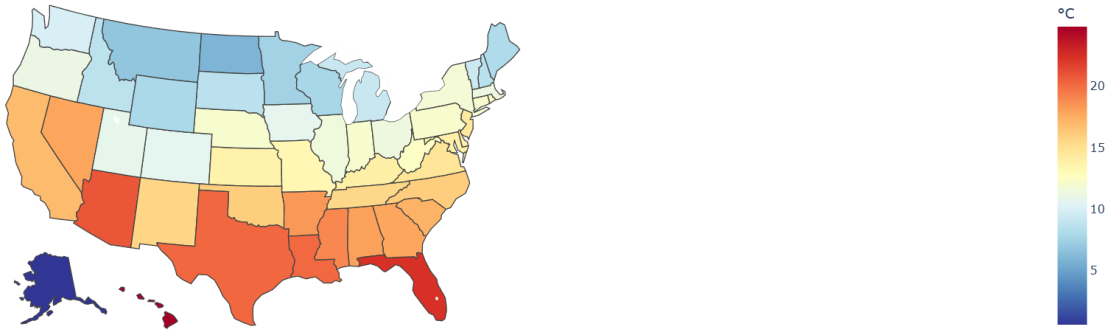


Figure 13: Average Ambient Temperature by State (2000–2016)

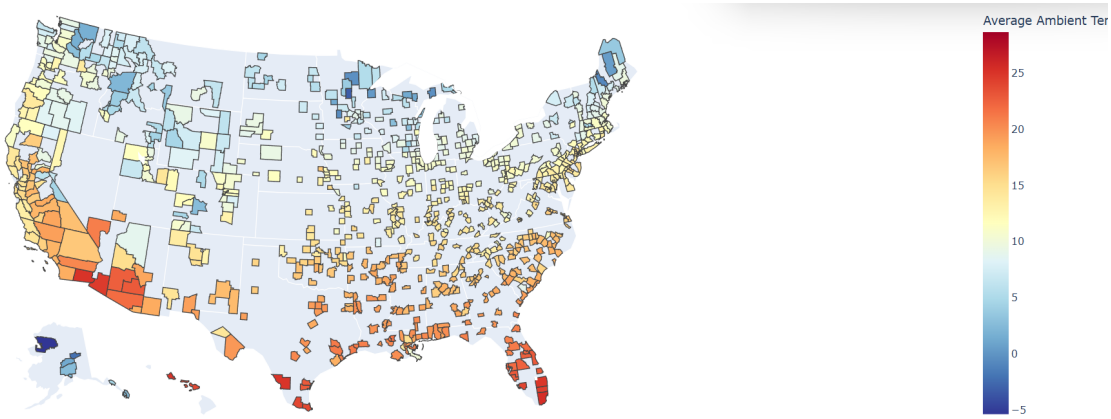


Figure 14: Average Ambient Temperature by County (2000–2016)

It is immediately apparent that southwestern states such as Arizona, Nevada, Texas, and parts of California exhibit the highest average ambient temperatures over the study period. In contrast, northern states such as Montana, North Dakota, Minnesota, and Maine display substantially lower annual averages. This geographic gradient is consistent with established climatic zones across the United States.

At the county level, variation becomes more pronounced. Desert counties in Arizona and Nevada consistently show some of the highest average temperatures in the nation, while mountainous and northern counties exhibit significantly lower averages. These findings are particularly important when considered alongside ozone concentrations, as higher temperatures promote ozone formation through enhanced photochemical reactions. The spatial overlap between high-temperature regions and elevated ozone concentrations supports this well-documented atmospheric relationship.

4.3.2 Wind Speed by State and County

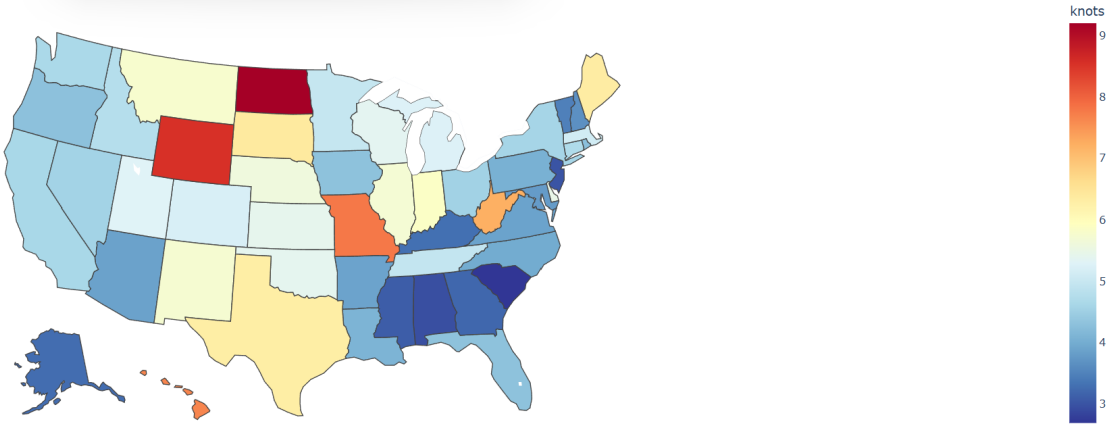


Figure 15: Average Wind Speed by State (2000–2016)

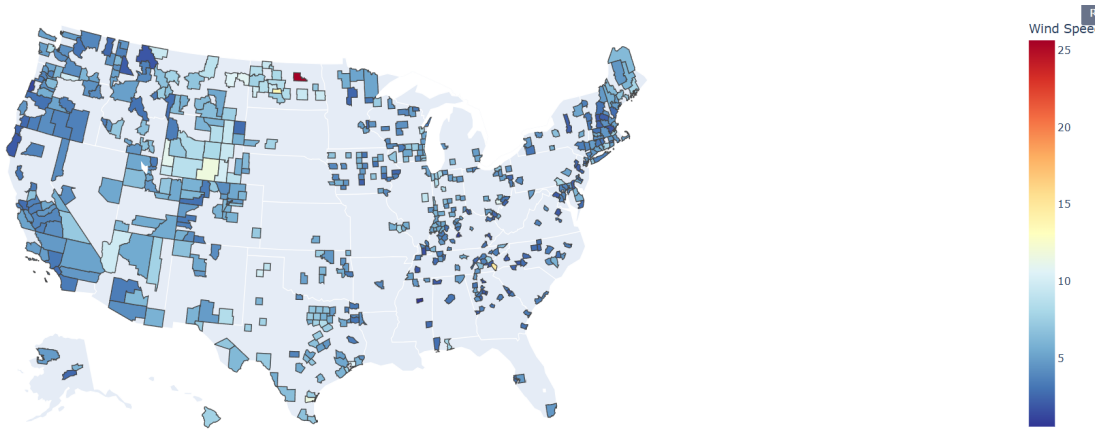


Figure 16: Average Wind Speed by County (2000–2016)

The data indicate that Great Plains and Midwestern states generally exhibit higher average wind speeds compared to coastal urban regions and geographically enclosed western counties. This pattern may help explain why certain industrial or agricultural states do not consistently exhibit the highest pollutant concentrations despite ongoing emissions activity.

In contrast, metropolitan regions characterized by lower wind circulation patterns may experience pollutant buildup, particularly for NO2 and PM2.5. Reduced atmospheric mixing increases the likelihood of prolonged exposure in densely populated areas, thereby increasing potential health risks.

4.3.3 Atmospheric Pressure by State and County

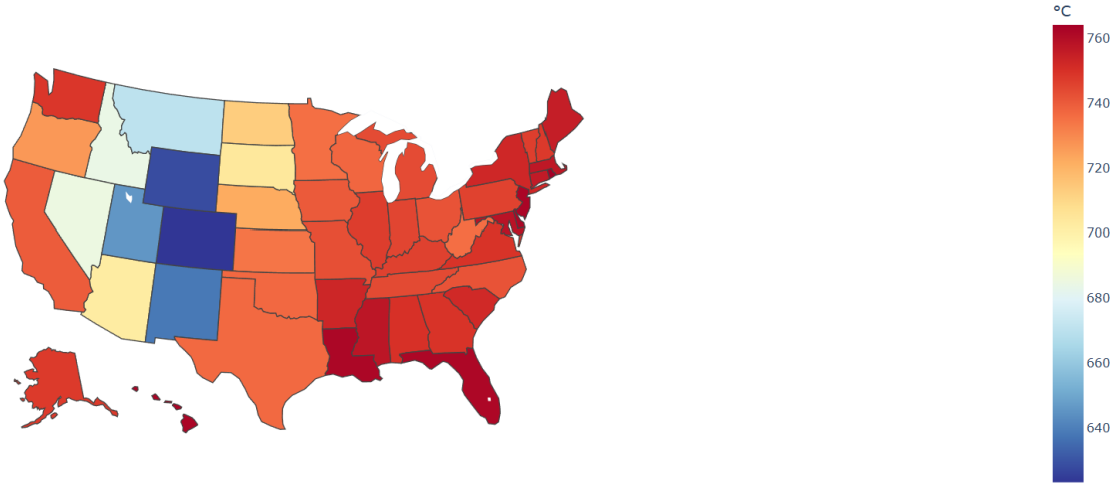


Figure 17: Average Atmospheric Pressure by State (2000–2016)

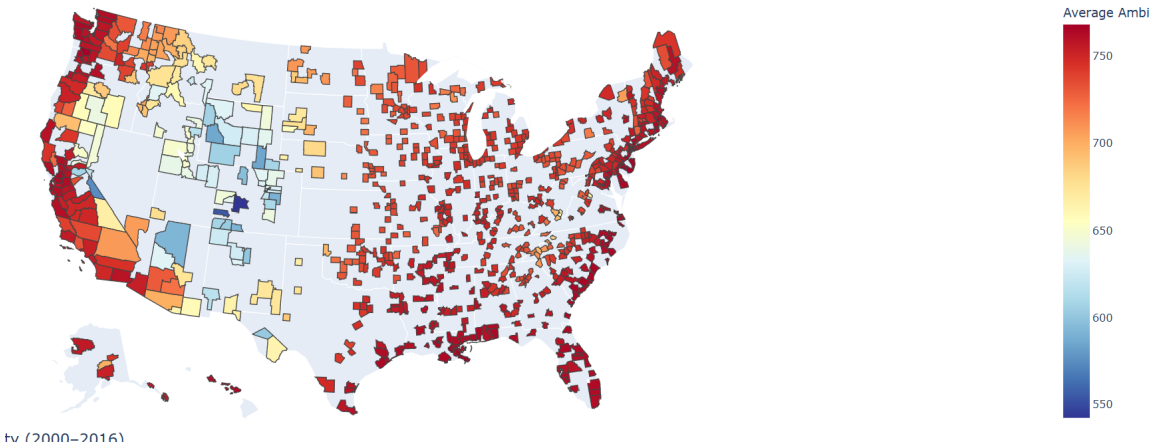


Figure 18: Average Atmospheric Pressure by County (2000–2016)

While atmospheric pressure does not vary as dramatically geographically as temperature, certain western and northern states exhibit seasonal conditions consistent with inversion-prone environments. These atmospheric stability conditions may help explain elevated NO2 and PM2.5 concentrations in some lower-population regions, as trapped air masses limit pollutant dispersion.

4.4 Respiratory-Related Crude Mortality Rates by State and County



Figure 19: Average Respiratory-Related Crude Mortality Rate by State from 2000-2016

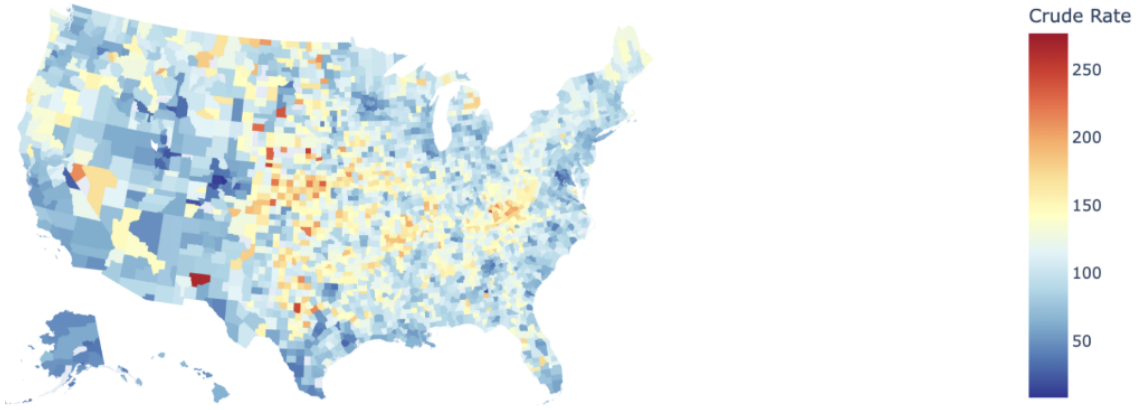


Figure 20: Average Respiratory-Related Crude Mortality Rate by County from 2000-2016

In Figures 19 and 20, respiratory-related crude mortality rates seem to be the highest in the Midwest and Southeast regions of the United States. For the most part, this tracks relatively well with our previous observations about vegetation loss, climate conditions, and especially air pollution. The crude rates are especially high West Virginia where there are somewhat high ozone concentrations and very high PM2.5 and NO2 concentrations. The South and parts of the Midwest also have higher levels of vegetation loss compared with most of the country. It is also worth noting that Alaska has the lowest respiratory-related crude rates. This makes sense considering that air pollutant concentrations in Alaska are not nearly as high as many of the Southern and Midwest countries, despite it having relatively high levels of vegetation loss. Moreover, average atmospheric pressure and ambient temperature are particularly higher in South and Midwest than other regions, which gives us insight into how these climate conditions may be correlated with crude mortality rates in those regions.

5 Machine Learning Results: Random Forest vs. Gradient Boosting Algorithms

5.1 Regional Analysis

5.1.1 Northeast

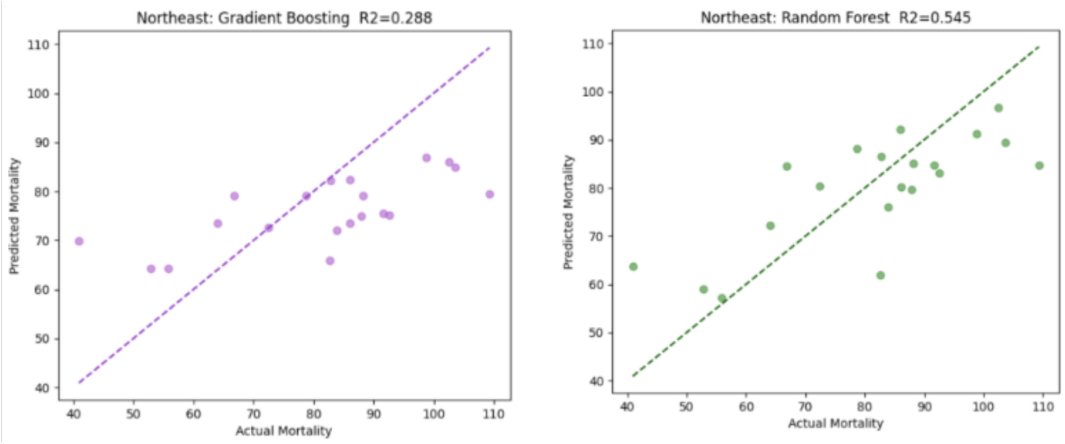


Figure 21: Northeast Model Comparison

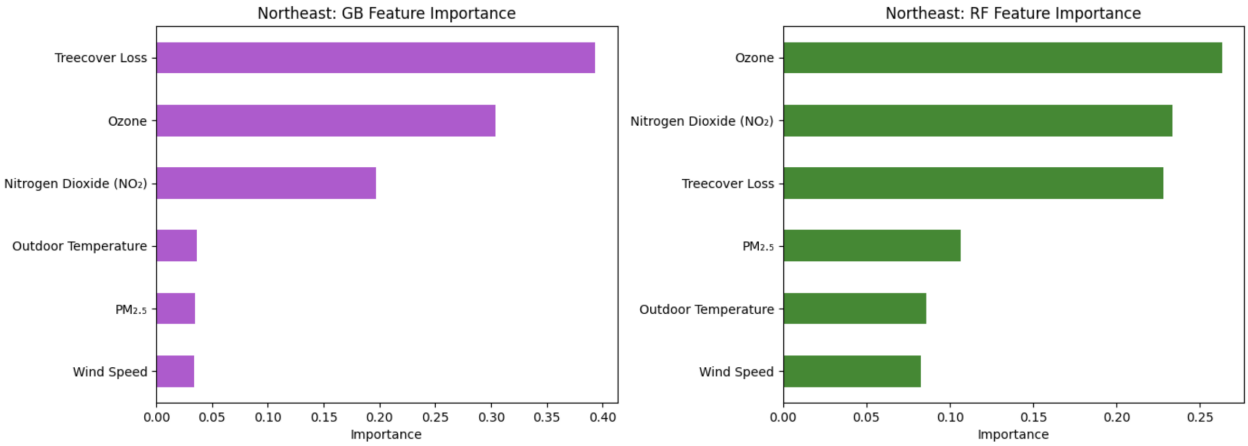


Figure 22: Northeast Feature Importance Comparison

Performance Metrics	Gradient Boosting	Random Forest
R^2	0.288	0.545
R^2 (Adjusted)	0.2401	0.5142
MAE	10.9278	9.1458
MAPE	10.9278	9.1457
MedAE	5.5125	6.9484
RMSE	14.3622	12.8051

Figure 23: Northeast Performance Metrics Comparison

When looking at the Northeast, Figures 21, 22, and 23 indicate that Random Forest performed better. In Figure 21, the R^2 value is more significant than the gradient boosting, and comparing that to the other metrics shown in Figure 23, it is safe to assume that Random Forest performed better. Moreover, in Figure 22, both models prioritized different features. The Gradient Boosting model's most important feature was tree cover loss and the Random Forest Model's most important feature was ozone concentration.

5.1.2 Midwest

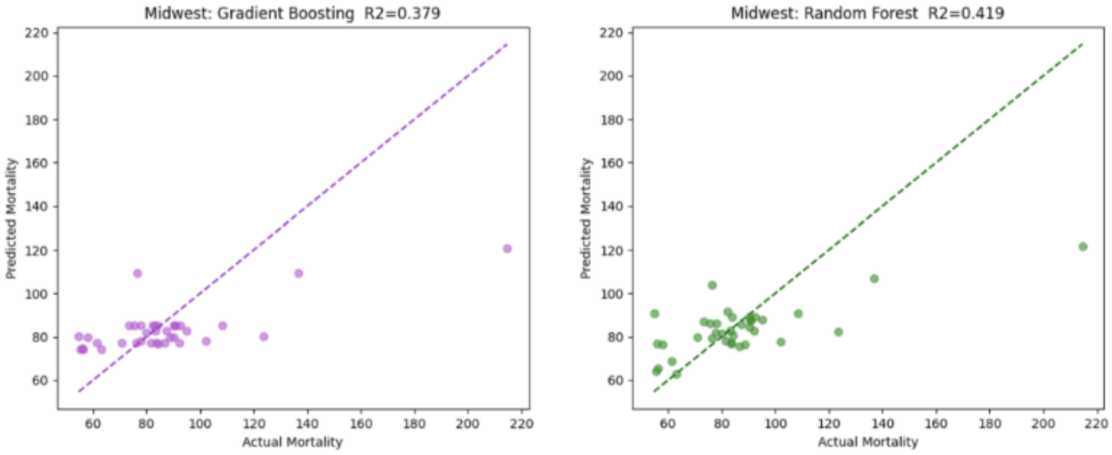


Figure 24: Midwest Model Comparison

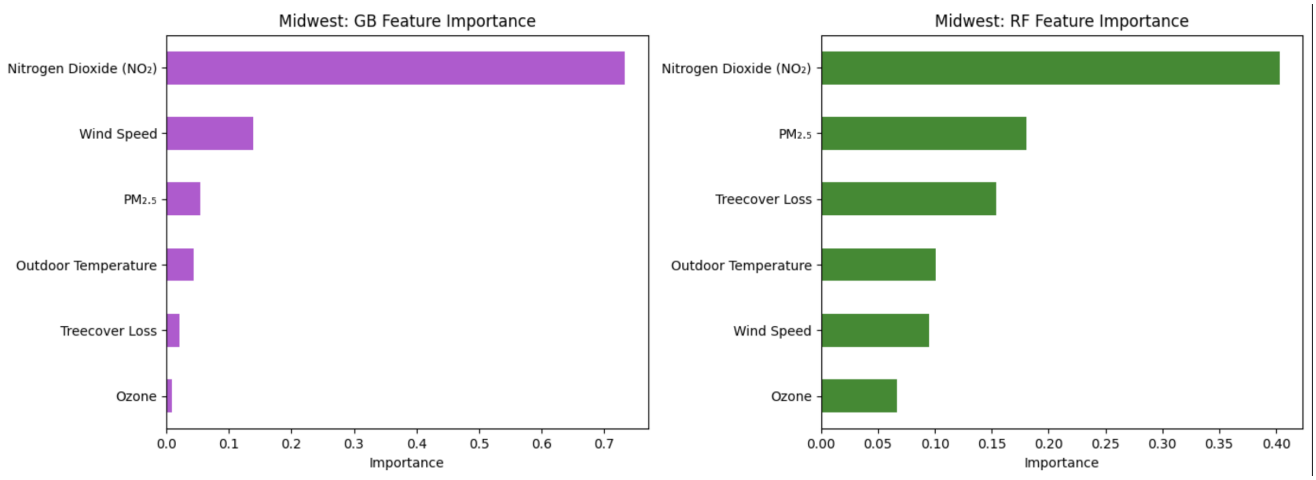


Figure 25: Midwest Feature Importance Comparison

Performance Metrics	Gradient Boosting	Random Forest
R^2	0.379	0.419
R^2 (Adjusted)	0.3574	0.3988
MAE	15.1393	12.2391
MAPE	15.1393	12.2391
MedAE	10.6321	10.0779
RMSE	20.1889	14.8257

Figure 26: Midwest Performance Metrics Comparison

When looking at the Midwest region, Figures 24, 25, and 26 indicate that Random Forest performed better. In Figure 24, the R^2 value is more significant than the gradient boosting, and comparing that to the other metrics shown in Figure 25 (especially MAE and MAPE), it is safe to assume that Random Forest performed better. Moreover, in Figure 26, both models prioritized the same feature: NO2.

5.1.3 South

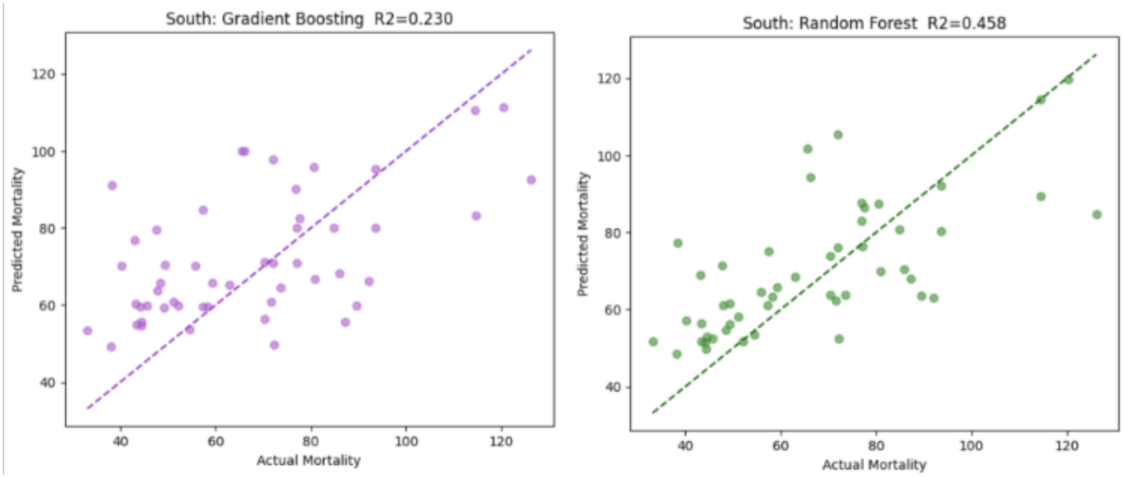


Figure 27: South Model Comparison

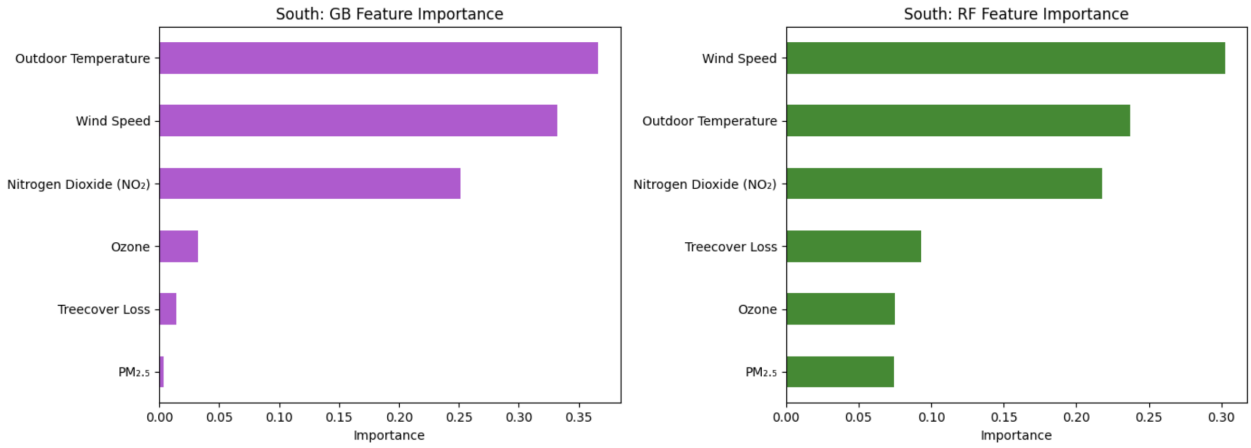


Figure 28: South Feature Importance Comparison

Performance Metrics	Gradient Boosting	Random Forest
R^2	0.230	0.458
R^2 (Adjusted)	0.211	0.444
MAE	14.7951	13.0014
MAPE	14.795	13.0014
MedAE	11.6623	7.871
RMSE	19.6715	17.8509

Figure 29: South Performance Metrics Comparison

When looking at the South, Figures 27, 28, and 29 indicate that Random Forest performed better. In Figure 27, the R^2 value is more significant than the gradient boosting, and comparing that to the other metrics shown in Figure 29, it is safe to assume that Random Forest performed better. Shifting focus to Figure 28, both models highlight NO2 as a key predictor, suggesting that traffic-related emissions and atmospheric conditions play a significant role in shaping respiratory risk in this region.

5.1.4 West

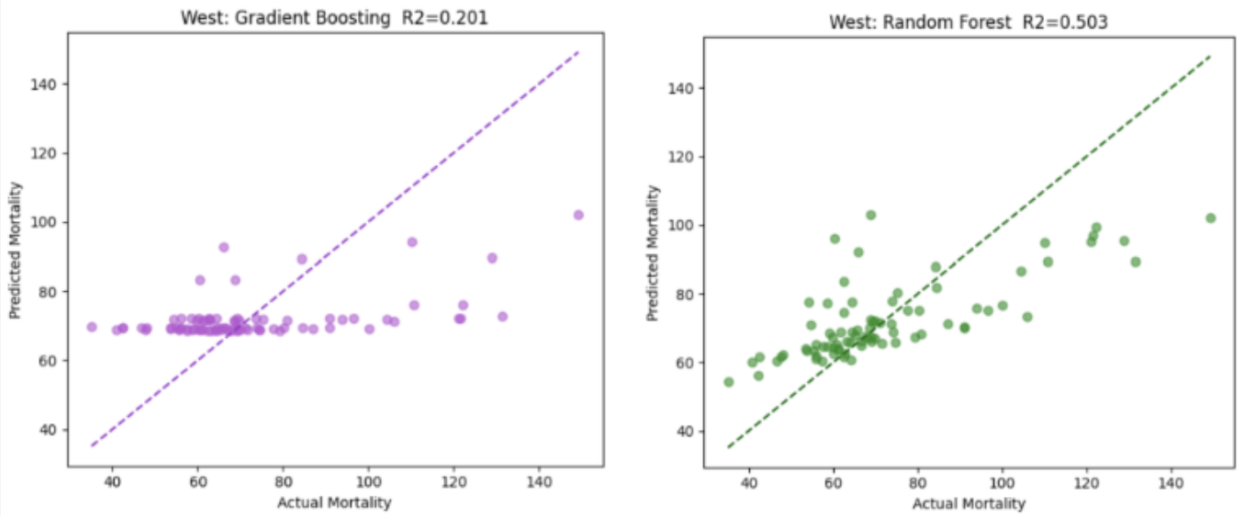


Figure 30: West Model Comparison

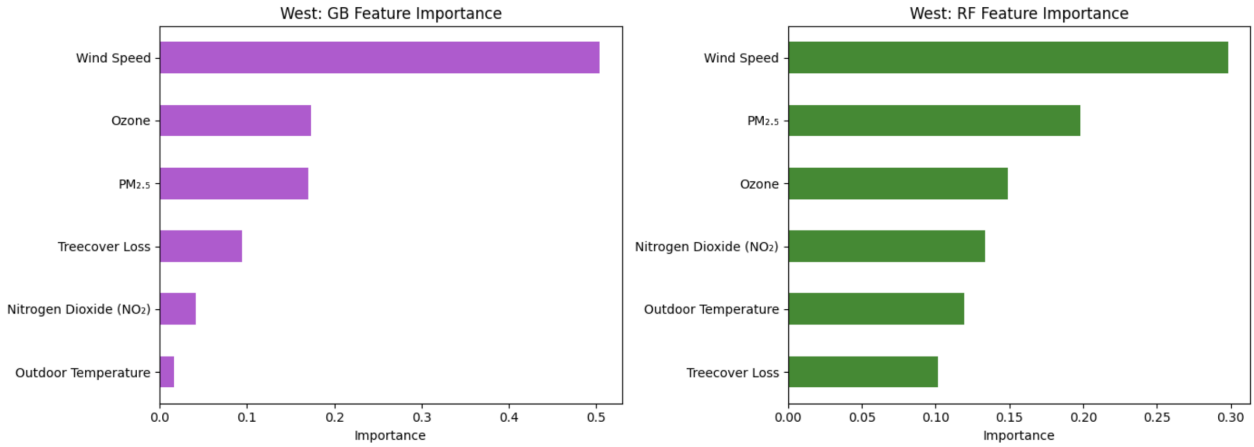


Figure 31: West Feature Importance Comparison

Performance Metrics	Gradient Boosting	Random Forest
R^2	0.201	0.503
R^2 (Adjusted)	0.1891	0.495
MAE	12.1393	11.1107
MAPE	12.1392	11.1106
MedAE	9.2382	7.2932
RMSE	16.4684	15.534

Figure 32: West Performance Metrics Comparison

Finally, looking at the Western region, Figures 30, 31, and 32 also indicate that Random Forest performed better. In Figures 30 and 32, the R^2 values and the other metrics perform better in the Random Forest model than the Gradient Boosting model. Looking at Figure 31, both models highlight wind speed as a key predictor, indicating that atmospheric conditions play a significant role in shaping respiratory risk in this region

5.2 Overall: The United States

5.2.1 Gradient Boosting Algorithm and Feature Importance

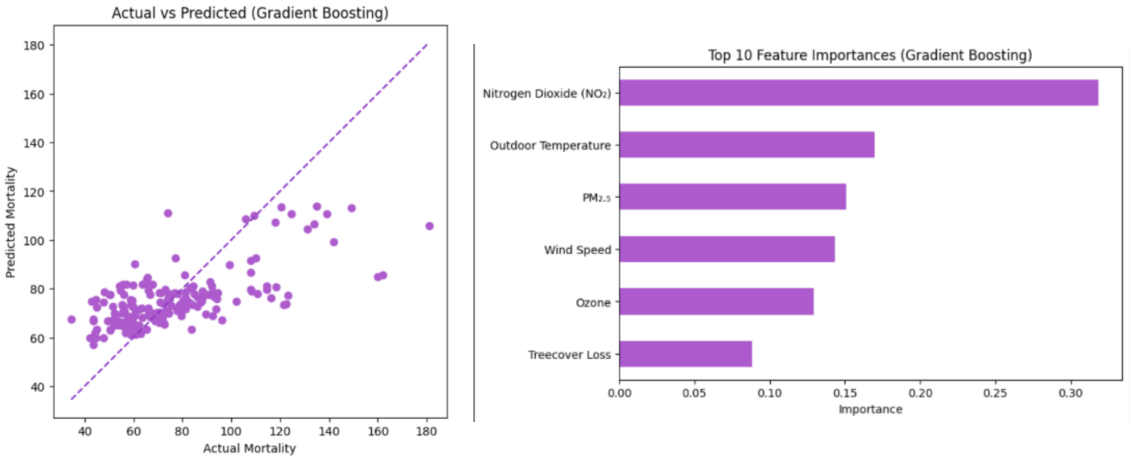


Figure 33: Gradient Boosting Algorithm and Feature Importance as Applied to the United States

Figure 33 presents the comparison between actual and predicted mortality values through the Gradient Boosting algorithm. We see a concentrated clustering of points in the lower ranges of mortality rates with a wider spread of points in the higher ranges. It is clear that the model is more sensitive to noise and variability, especially when compared to the Random Forest model below. Moreover, the most influential predictor in the Gradient Boosting algorithm, as applied to the whole United States, was NO2.

5.2.2 Random Forest and Feature Importance

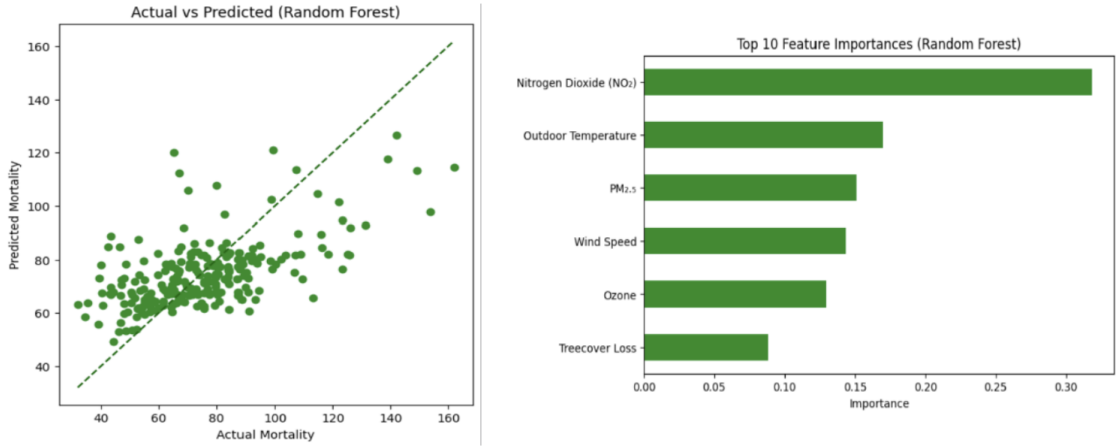


Figure 34: Random Forest Model and Feature Importance as Applied to the United States

Figure 34 presents the comparison between actual and predicted mortality values. The clustering of points around the diagonal indicates that the model captures overall trends, though noticeable dispersion reflects prediction error for certain counties. Further, feature importance analysis reveals that nitrogen dioxide (NO2) is the most influential predictor, followed by temperature-related variables, ozone, wind speed, and PM2.5. The prominence of NO2 is consistent with the correlation analysis, where it exhibited a strong (though negative) relationship with mortality. This suggests that NO2 may be capturing broader structural factors such as urbanization rather than acting solely as a direct pollutant indicator.

6 Risk Classification Results

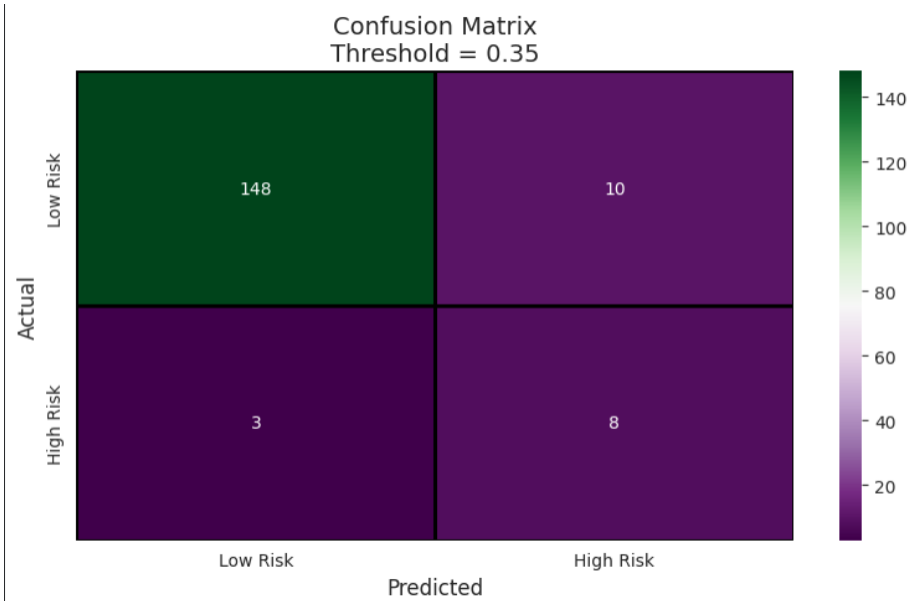


Figure 35: Risk Classification Confusion Matrix

The confusion matrix (Figure 31) shows that the model correctly classified the majority of counties, achieving an overall accuracy of 0.92. Specifically, 148 low-risk counties and 8 high-risk counties were correctly justified, while 10 low-risk counties were misclassified as high-risk and 3 high-risk counties were misclassified as low risk.

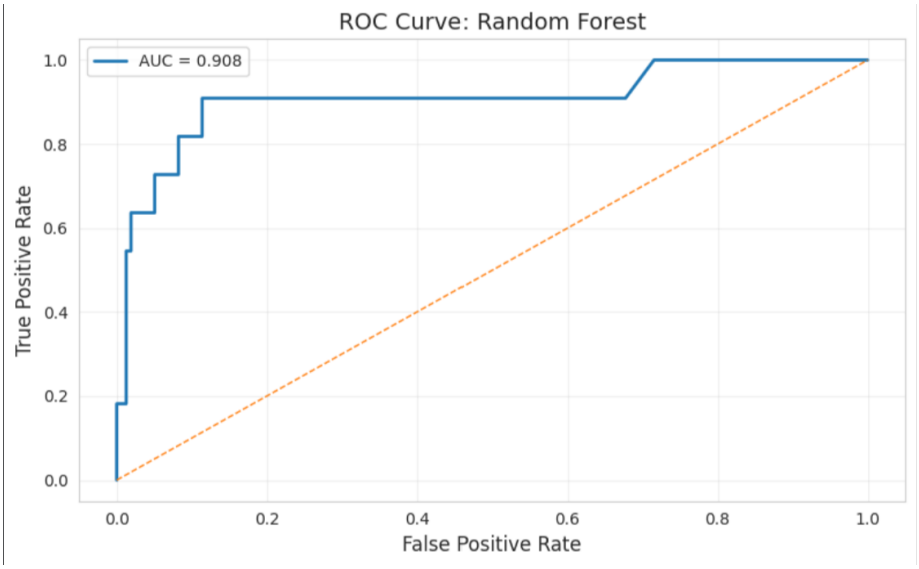


Figure 36: Risk Classification ROC Curve

The ROC curve (Figure 32) further demonstrates strong model performance with a curve that closely approaches the top-left corner, reflecting a favorable balance between true and false positive rates.

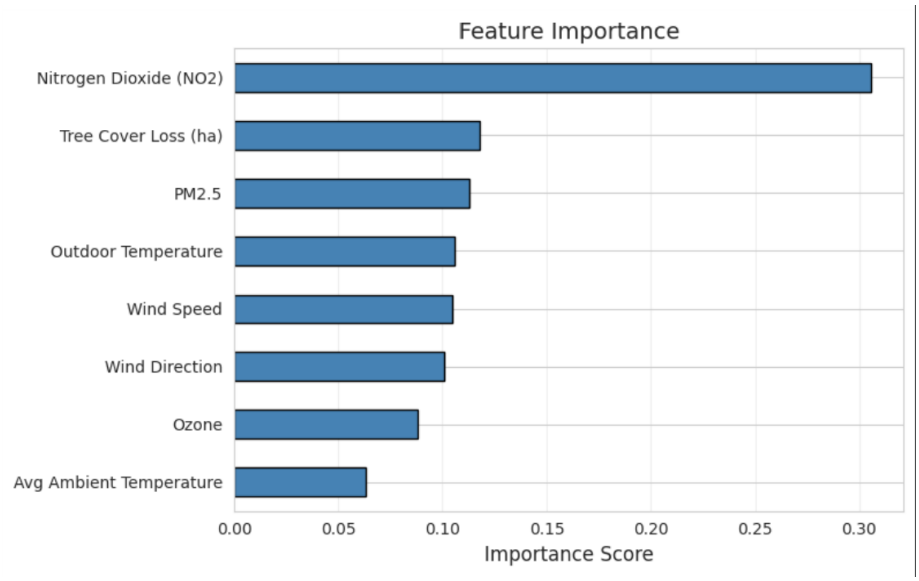


Figure 37: Risk Classification Feature Importance

Feature importance analysis (Figure 33) indicates that nitrogen dioxide is the most influential predictor, followed by tree coverage loss, PM2.5, and temperature-related variables.

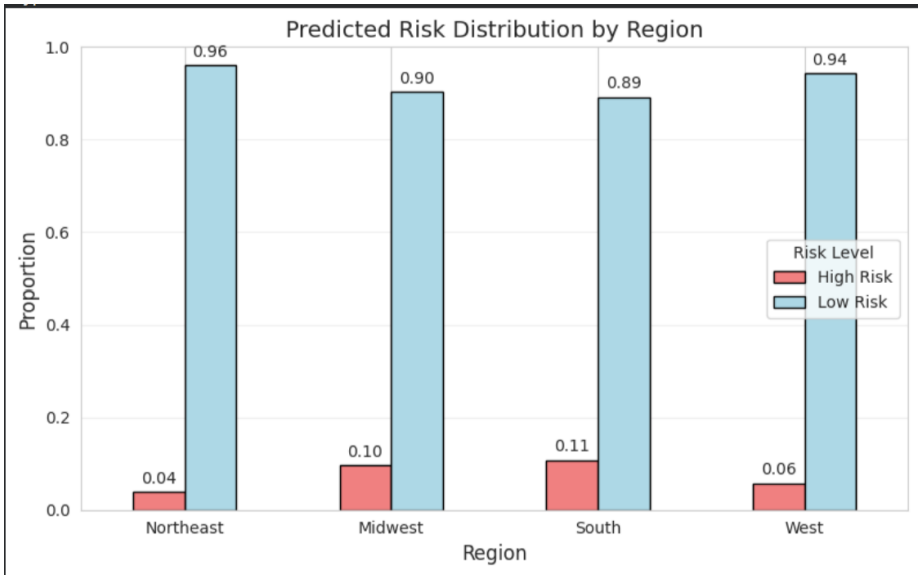


Figure 38: Risk Classification Regional Distributions

The regional analysis (Figure 34) shows variation in predicted risk across U.S regions. The South and Midwest exhibit higher proportions of predicted high-risk counties compared to the West and Northeast.

7 Discussion

7.1 EDA

As we previously surveyed different environmental factors across the United States, we will now evaluate those results against respiratory-related crude mortality rates to support our subsequent findings from our machine learning and risk classification models.

First, we will discuss air pollution. According to the heat map of NO2 concentrations in the United States we discussed earlier, we see the highest concentrations of NO2 in parts of the Midwest and most of the Northeast. Some counties in Arizona and California, which are densely populated, also show high levels of NO2. Moreover, the heat map of PM2.5 concentrations show the highest levels in the South and Midwest regions as well as a few densely populated counties in California. Finally, O3 concentrations are highest in the western states, like Utah, Arizona, Colorado, and parts of California. It is also worth noting that the ozone concentrations in the South and Midwest regions are not at all insubstantial. Taking all our observations about air pollution together, we arrive at a clearer understanding of the crude mortality rate heat map. We see the highest mortality rates in the Midwest and parts of the South, specifically West Virginia, where PM2.5 and NO2 concentrations are notably high. It is also worth addressing the relatively low crude mortality rates near the West Coast, namely California and Arizona. The air pollutant concentrations in those States and the counties therein are noticeably high, yet the crude mortality rates are relatively low. This apparent discrepancy might be explained by the lack of sufficient county data in those regions or, more likely, the effects of confounding variables like climate and vegetation density. Nevertheless, from this observational analysis of the heat maps, we derived meaningful insight into the regional relationships between air pollution and respiratory-related crude mortality rates in the United States.

We will now move on to a discussion about vegetation density in the United States to help better explain the mortality trends we observed. In 2001 and 2016, the loss of tree coverage was highest in the South and Northwest regions, with an interesting outlier observation in Maine and Alaska.

Most notably, however, was the substantial increase in vegetation loss in California, the Northwest, and Alaska since 2001, as observed in the 2016 heat map. We hypothesized that the loss of tree coverage would support an unhealthy environment prone to respiratory-related health issues, which in turn could lead to death. Our data seems to support our hypothesis, given the relatively high crude mortality rates in parts of northern California, the South, and Maine. Our tree-coverage-based observations are, by no means, a perfect reflection of respiratory-related mortality, and we certainly would not expect it to be perfect. We are merely recognizing the presence of a non-trivial correlative relationship between vegetation loss and respiratory-related mortality rate. Notably, most of the United States has not suffered from a substantial loss of tree-coverage, the regions of which overlap with where we observed the highest crude mortality rates. Thus, just because there was no substantial vegetation loss in a particular region, it does not follow that that region would not be at high risk of respiratory-related mortality. There are always more variables to consider.

Finally, we will survey our climate conditions factors as it relates to respiratory-related crude mortality rates to complete our preliminary analysis and arrive at a more comprehensive conclusion about our overall observations. From the ambient temperature heat map, it is clear that from coast to coast, the southern half of the United States has higher temperatures than the northern half, which of course is consistent with general temperature trends in the United States. We do not, however, see such segmented regional trends in wind speed by state and county, but there are still some important observations to note despite the substantial lack of county data. We see the highest relative wind speeds in the Midwest region and, notably, West Virginia, whereas the lowest wind speeds reside in the South and parts of the Northeast. Finally, the atmospheric pressure heat map reveals a sharp regional contrast. The eastern part of the United States has the highest atmospheric pressure, while many states in the West, excluding the coastal states (California, Oregon, and Washington), have relatively low levels of atmospheric pressure. Taking all of these climatic observations together, when comparing them side by side with respiratory-related crude mortality rates, we see slightly similar geographic trends. The eastern part of the United States, particularly the South and Midwest where atmospheric pressure is high, wind speeds are low, and temperatures are relatively warmer, is replete with higher mortality rates. Concurrently, the non-coastal regions of the West, where atmospheric pressure is low and ambient temperatures are high, we see lower crude mortality rates. These climatic trends, of course, are not determinative on respiratory-related mortality, but they certainly capture the overall mortality trends we observed, especially when taken together with air pollution and vegetation density variables.

From what can be gleaned from our exploratory data analysis above, we can see how different environmental factors may help contribute to an explanation about the general respiratory-related mortality trends we see across the United States. Air pollution concentrations are strongest in the South and Midwest, with a few outlier regions in California and New England. Vegetation density loss is most prevalent in the South and Northwest regions of the United States, where we would in turn expect higher mortality rates. Moreover, the various climate variables we observed interestingly coincided with the general regional mortality trends, but our observations were by no means a perfect reflection of those trends. Ultimately, our analysis here importantly contextualizes the bulk of our further regional analysis through machine learning and risk classification models. Next, we will discuss those models in detail and reflect on their implications.

7.2 Machine Learning Models and Performance Comparison

To evaluate predictive performance, Gradient Boosting and Random Forest models were applied across four U.S regions: Northeast, Midwest, South, and West. The performance of the model was assessed using R^2, adjusted R^2, Mean Absolute Error(MAE), mean absolute probability Error(MAPE), and Median Absolute Error(MedAE). Across all regions, the Random Forest model consistently outperformed Gradient Boosting in terms of predictive accuracy and error reduction

In the Northeast region of the United States, both models reveal relatively strong predictive capabilities; however, Random Forest consistently outperforms Gradient Boosting across the ma-

jority of the metrics. Random Forest achieves a higher test R^2 of 0.6647, in contrast to 0.5782 for Gradient Boosting, indicating that it accounts for a greater proportion of variance in mortality outcomes on unseen data. The higher training R^2 values (0.8089 for Random Forest and 0.7053 for Gradient Boosting) suggest that both models fit the training data well, yet the difference between training and test performance indicates a degree of overfitting, particularly for Random Forest. Error metrics further validate Random Forest's advantage, with lower MAE (9.1458 compared to 10.9276), MAPE(9.15% versus 10.93 %), and RMSE (12.81 against 14.36), indicating more accurate and consistent predictions overall, although Gradient Boosting achieves a slightly lower median absolute error (5.51 versus 6.95), suggesting fewer extreme individual errors. The feature importance results indicate that ozone and tree cover loss are the primary predictors in this area, with nitrogen dioxide (NO2) also contributing significantly. The prominence of ozone suggests that photochemical pollution plays a crucial role in respiratory outcomes in the Northeast, while the influence of vegetation loss underscores the impact of environmental degradation. Random Forest's ability to incorporate multiple contributing variables likely explains its superior performance compared to Gradient Boosting, which places greater emphasis on a limited number of dominant features.

In the Midwest, the Random Forest model clearly outperforms the Gradient Boosting model in every performance metric. In our preliminary ML analysis, we learned that the Random Forest model had a substantially larger train R^2 value (0.7411) than the Gradient Boosting model (0.1012). Moreover, the Random Forest test R^2 value (0.523), while lower than its train value, was significantly larger than the Gradient Boosting test R^2 value (0.1154). The other metrics additionally supported the finding that the Random Forest model was superior. The mean absolute error and mean absolute percentage error were both lower in the Random Forest model (12.2391 compared to 15.1303 for Gradient Boosting). Moreover, the root mean squared error (RMSE) was lower for the Random Forest Model (14.8257 compared to 20.1889). Taking all of these metrics together, it is abundantly clear that the Random Forest model predicts respiratory-related crude mortality rates in the US better than the Gradient Boosting model, exhibiting much lower error rates. It is also worth noting that nitrogen dioxide contributed most significantly to both models' predictive analysis. The primary difference, though, is the larger NO2 proportion in the Gradient Boosting model, whereas the Random Forest model more heavily incorporated other features like PM2.5 and tree cover loss. Again, the Random Forest model's ability to incorporate multiple contributing variables may explain its superior performance compared to Gradient Boosting, which places greater emphasis on a limited number of dominant features.

In the Southern region, both models show predictive effectiveness, with Random Forest once again exceeding Gradient Boosting. Random Forest achieves a higher test R^2 of 0.5109 compared to 0.4060, which indicates a stronger capacity to account for variability in mortality outcomes. The training R^2 values (0.7952 for RF and 0.6203 for GB) suggest that both models fit the training data reasonably well; however, the differences between training and test performance indicate some overfitting. Error metrics consistently favor Random Forest, with a lower MAE (13.0014 versus 14.7951), MAPE (13% against 14.8%), and RMSE (17.85 compared to 19.67), reflecting improved prediction accuracy and reduced error magnitude. Additionally, Random Forest achieves a notably lower median absolute error (7.87 compared to 11.66), indicating greater strength against outliers. Feature importance analysis shows that both nitrogen dioxide (NO2) and wind speed are significant predictors in the South, with temperature and vegetation-related variables also contributing. The importance of wind-related variables suggests that atmospheric transport and pollutant dispersion are critical factors in determining respiratory risk in this region. Random Forest's ability to incorporate these multiple interacting environmental factors likely enhances its performance, while Gradient Boosting's more concentrated feature weighting may limit its capacity to fully capture these dynamics.

In the West, the Random Forest model outperforms the Gradient Boosting model in every performance metric, but the differences are less substantial than, say, the Midwest models. In our preliminary ML analysis, we learned that the Random Forest model had a larger train R^2

value (0.6878) than the Gradient Boosting model (0.4469). Moreover, the Random Forest test R^2 value (0.4772), while lower than its train value, was slightly larger than the Gradient Boosting test R^2 value (0.4125). The other metrics additionally supported the finding that the Random Forest model was superior. The mean absolute error and mean absolute percentage error were both lower in the Random Forest model (11.1107 compared to 12.1393 for Gradient Boosting), but the difference here was relatively insubstantial. Further, the median absolute error for the Random Forest model was lower (7.2932 compared to 9.2382). Moreover, the root mean squared error (RMSE) was lower for the Random Forest Model (15.534 compared to 16.4684). Taking all of these metrics together, it is abundantly clear that the Random Forest model predicts respiratory-related crude mortality rates in the US better than the Gradient Boosting model, exhibiting much lower error rates. It is also worth noting that wind speed (one of our climate variables) contributed most significantly to both models' predictive analysis. The Random Forest model, however, more meaningfully incorporates other important features like PM2.5, Ozone, and NO2 (all air pollutants). Once again, the Random Forest model's ability to incorporate multiple contributing variables might explain its superior performance compared to Gradient Boosting, which places greater emphasis on a limited number of dominant features.

7.3 Risk Classification

The results of the risk classification model demonstrate strong predictive performance and reveal meaningful patterns in respiratory health risk across the United States.

The model demonstrated a high AUC of 0.908, which reflects a strong capacity to differentiate between high- and low-risk counties across a spectrum of classification thresholds. Additionally, the overall accuracy of 0.92 implies that the model is effective in accurately classifying counties in general. However, given the dataset's imbalanced characteristics, accuracy alone is insufficient for evaluating the model's performance.

More significantly, the model achieved a recall of 0.73 for the high-risk class, indicating that roughly 73% of high-risk counties were correctly identified. This outcome is crucial, as overlooking high-risk counties (false negatives) would diminish the model's relevance in a public health framework. The confusion matrix further substantiates this, indicating that only 3 high-risk counties were misclassified as low risk.

On the other hand, precision for the high-risk class was lower at 0.44, suggesting that a segment of counties predicted as high risk were actually low risk. This tradeoff between precision and recall is expected, particularly in light of the use of SMOTE and the priority given to identifying high-risk regions. The corresponding F1-score of 0.55 illustrates this balance, capturing both the model's capability to detect high-risk counties and the incidence of false positives.

Geographically, the model identifies the South and Midwest as regions with the highest proportions of high-risk counties, with approximately 11% and 10% of counties deemed high risk, respectively. In contrast, the Northeast has the lowest proportion (around 4%). These findings are in line with established regional disparities in environmental conditions and health outcomes, where southern regions often face higher pollution levels and increased susceptibility to respiratory diseases.

Results regarding feature importance further highlight the significance of environmental factors. Nitrogen dioxide (NO2) emerged as the most critical predictor, with an importance score of approximately 0.31, significantly surpassing all other variables. This indicates that traffic-related air pollution is a major factor influencing respiratory risk. Additional contributors, such as PM2.5 and the loss of tree cover, further underscore the combined effects of air quality and environmental degradation on health outcomes.

8 Conclusions

This analysis examined how air pollution, vegetation loss, and climatic conditions interact to influence respiratory-related mortality across the United States. By integrating environmental and health data sets and applying both exploratory analysis and machine learning models, this research provides a comprehensive, regionally grounded understanding of respiratory health risk.

In response to the central question, the results indicate that respiratory health outcomes are influenced by the interaction of multiple environmental factors rather than any single variable. Air pollutants, particularly nitrogen dioxide (NO2), PM2.5, and ozone, consistently emerged as key contributors, while the loss of vegetation and climate variables such as temperature and wind further shaped regional exposure patterns. These findings reinforce the significance of viewing environmental health within a dynamic system where pollution, climate, and land conditions interact to affect respiratory outcomes.

To address the sub-question, the results demonstrate that machine learning models can effectively predict respiratory mortality using environmental variables, with Random Forest consistently outperforming Gradient Boosting across all regions in terms of R^2 and error metrics, indicating a stronger predictive capability and generalization. Feature importance analysis further reveals that nitrogen dioxide (NO2) is the most influential predictor across both regression and classification models, with additional contributions from PM2.5, ozone, vegetation loss, and climate variables such as temperature and wind, highlighting the combined role of air quality and environmental context. The improved performance of machine learning models compared to simple correlation analysis also suggests the presence of nonlinear relationships and interactions between variables, as patterns that appeared weak or counterintuitive in the exploratory analysis were more effectively captured in multivariate models. Finally, the risk classification model successfully identified high- and low-risk counties, achieving strong performance (AUC = 0.908, accuracy = 0.92) and revealing clear regional disparities, with the South and Midwest exhibiting higher proportions of high-risk counties compared to the Northeast and West.

Despite these strengths, it is important to recognize several limitations. Firstly, the vegetation dataset focuses on tree cover loss instead of total vegetation density, which may not adequately reflect the protective benefits of green spaces. Secondly, the analysis is confined to counties with adequate data availability, which may inadvertently exclude rural or less-monitored areas, leading to potential sample bias. Furthermore, the models do not take into account socioeconomic status, healthcare access, or demographic variables, all of which are known to affect respiratory health outcomes. Lastly, although machine learning models demonstrate robust predictive capabilities, they do not establish causal links between environmental exposures and mortality.

Future research could build upon this analysis by integrating additional variables such as socioeconomic factors, healthcare systems, and demographic information to provide a more nuanced understanding of environmental risks. Incorporating more recent data and higher-resolution spatial data could enhance the accuracy and applicability of the models. Moreover, employing causal inference techniques or time-series methodologies may assist in unraveling the intricate relationships between environmental exposures and respiratory health outcomes. Increasing access to more detailed health datasets, including hospitalization records or disease-specific information, would also facilitate a more thorough evaluation of respiratory health beyond mere mortality statistics.

Ultimately, this research confirms that environmental conditions are essential in influencing respiratory health outcomes in the United States. By merging air pollution, vegetation, and climate factors within a machine learning framework, the study not only enhances predictive performance but also offers significant insights into regional disparities in respiratory risk. These findings stress the importance of integrated environmental management and focused public health strategies in addressing respiratory health issues.

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9 Code

All of the code can be found at this GitHub link.